



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

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SECR1033 COMPUTER ORGANISATION AND ARCHITECTURE

GROUP PROJECT

CPU & HARDWARE PERFORMANCE BENCHMARKS

SECTION 3

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Benchmarking Demonstration: <https://youtu.be/9z05hiuaKso>

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1.0 Introduction

Benchmarking is the act of running a set of operations in order to assess the relative performance of an object by running a number of standard tests and trials against it, and comparing it between different variants of the same object to find the best in value overall. CPU benchmarking is a process of running a series of tests and experiments that was designed to measure the performance of a computer or device CPU.

There are three different types of benchmarking, which are synthetic, real-world and application benchmark. **Synthetic benchmarking** involves running specially designed programs that simulate a range of tasks to measure a system's performance to have easily repeatable results for accurate comparisons and minimal bottlenecks across different isolated tests.. These benchmarks are created to stress specific components of the system, such as the CPU, GPU, or memory. For example, Geekbench is one of the benchmarking tools that uses synthetic benchmarking. While synthetic benchmarks provide consistent and repeatable results, they may not always reflect a user's actual experience with the machine since they do not replicate actual usage scenarios. Conversely, **Real-World Benchmarking** is a benchmarking technique that evaluates the system by giving it a real life workload, such as web browsing, file compression, 3D rendering to assess the performance of computers completing those tasks. For example, using Task Manager in Windows to monitor system performance while running actual tasks. Real-world benchmarking provides practical insights into system performance as a whole, but can be more variable and less controlled compared to synthetic benchmarks, making direct comparisons and assessment regarding the performance of individual components more challenging. Meanwhile, **Application Benchmarking** is a type of real-world benchmarking that is focused on that one specific software application to evaluate system performance in the particular application. Application benchmarks are relevant for users who rely on a certain software, by providing direct insights for those applications. These benchmarks consider factors like load times, execution speed, and responsiveness, offering a detailed view of performance in a specific context. For example, Winstone 97 that tests a PC's overall performance when running Windows-based 32-bit business applications.

This project is focused on benchmarking different Central Processing Units (CPUs) and hardware performance using free available tools. It was divided into two parts. In part 1, a CPU benchmarking and performance analysis will be carried out across different computers using free tools available. There are 3 benchmarking tools used in Part 1, which are CPU-Z, Geekbench, Task Manager and Minitools. The benchmarking process will be carried out across 3 different computers, which is ASUS VivoBook S14, Lenovo IdeaPad Flex 5, and Acer Nitro 5 Intel. Meanwhile in Part 2, the writing of an ASM code program as Benchmarking Software will be required. The data collected will then be compared and the performance of each computer will then be analysed and evaluated in the next stage of the project.

1.1 Problem Background

In the rapidly evolving world of technology nowadays, computers are widely used in our daily lives. There are all kinds of computers existing on the market, presenting a large variety of choices of computers with different specifications and compatibility, at the same time, making it hard for a user to choose from it while considering the tradeoff between space, time and price. This makes the ability to understand the performance capabilities of computers essential, and extremely useful for the people. This project focuses on benchmarking Central Processing Units (CPUs) and hardware performance using freely available tools. CPU Benchmarking is a series of tests to evaluate the performance of a computer or CPU, as well as to compare between different systems, with freely available CPU benchmark software.

1.2 Aim and Objective

The main goal of this grouped project is to equip students with the knowledge and skills to perform CPU benchmarking with freely available tools as well as offering insights into computer performance, hardware differences, and the intricacies of software optimization, while also enhance the student's ability of assembly language, as the objectives stated below:

1. To understand the principles of CPU benchmarking.
2. To compare the performance of different CPUs.
3. To analyse the impact of different workloads on CPU performance.
4. To learn to use free benchmarking tools and interpret their results.
5. To write a program in ASM language to measure and compare the execution time of different loop iterations to assess CPU performance under varying computational loads.

1.3 Scopes

The scope of this project encompasses a detailed examination of computer specifications and the features of benchmarking tools, ensuring a comprehensive understanding of both areas. The specifications of computer used in the benchmarking process are as below:

Specifications	Computer Type			
	Lenovo IdeaPad Flex 5	ASUS VivoBook S14	Acer Nitro 5 Intel	Acer Nitro AN515-55
CPU / Processor	Intel Core i5-1135G7	Intel Core i5-8250U	Intel Core i7-11800H	Intel Core i7-10750H
RAM	8GB	12GB	16GB	16GB

GPU	Intel Iris Xe Graphics	NVIDIA GeForce MX15	NVIDIA GeForce RTX 3050	NVIDIA GeForce RTX 3050
Storage	SSD 475GB	SSD 237GB	SSD 512GB	SSD 512GB
Operating System	Windows 11 Home	Windows 11 Home	Windows 11 Home	Window 11 Home

The tools used in the benchmarking process are:

1. CPU-Z

CPU-Z is used to collect detailed information and specifications of the CPU and hardware components, such as:

- CPU (Processor name, Codename, Clock)
- Mainboard (Motherboard, Chipset, BIOS)
- Memory (Type, Size, Timings)

2. Geekbench

Geekbench is used to add system info as the complement to CPU-Z, including system information, CPU information and memory information. Geekbench conducts CPU tests with a bunch of short tasks with periods of cooldown time between them, to test the burst performance of both single core and multi-core performance and generate performance points. Their online browser allows users to see individual microbenchmarks and the comparative analysis of different CPUs under various conditions.

3. Monitor Task Manager (Windows OS tools)

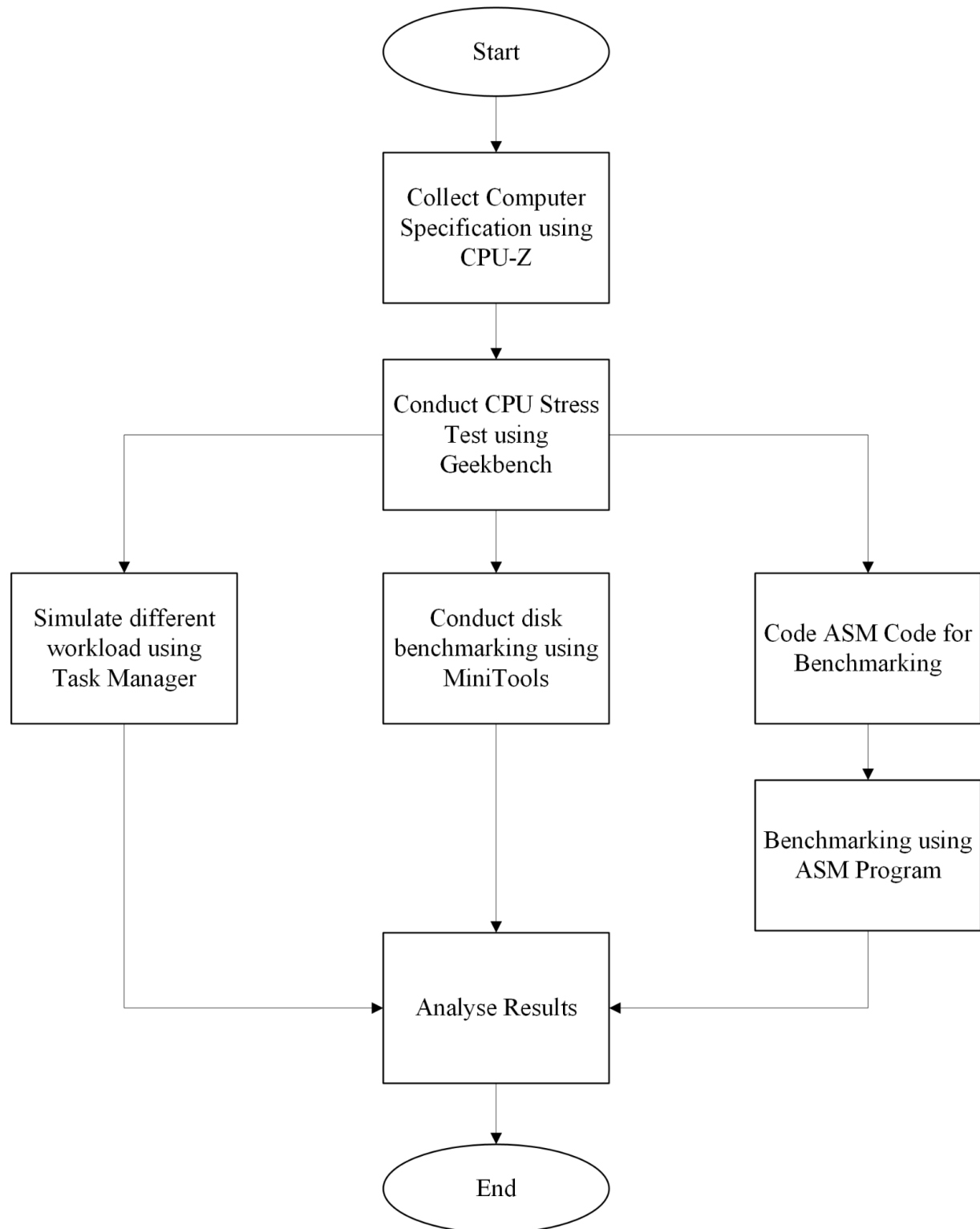
Task Manager by Windows is a tool to carry out Real-World Benchmarking for assessing performance of computers under real life workload.

4. MiniTool Partition Wizard

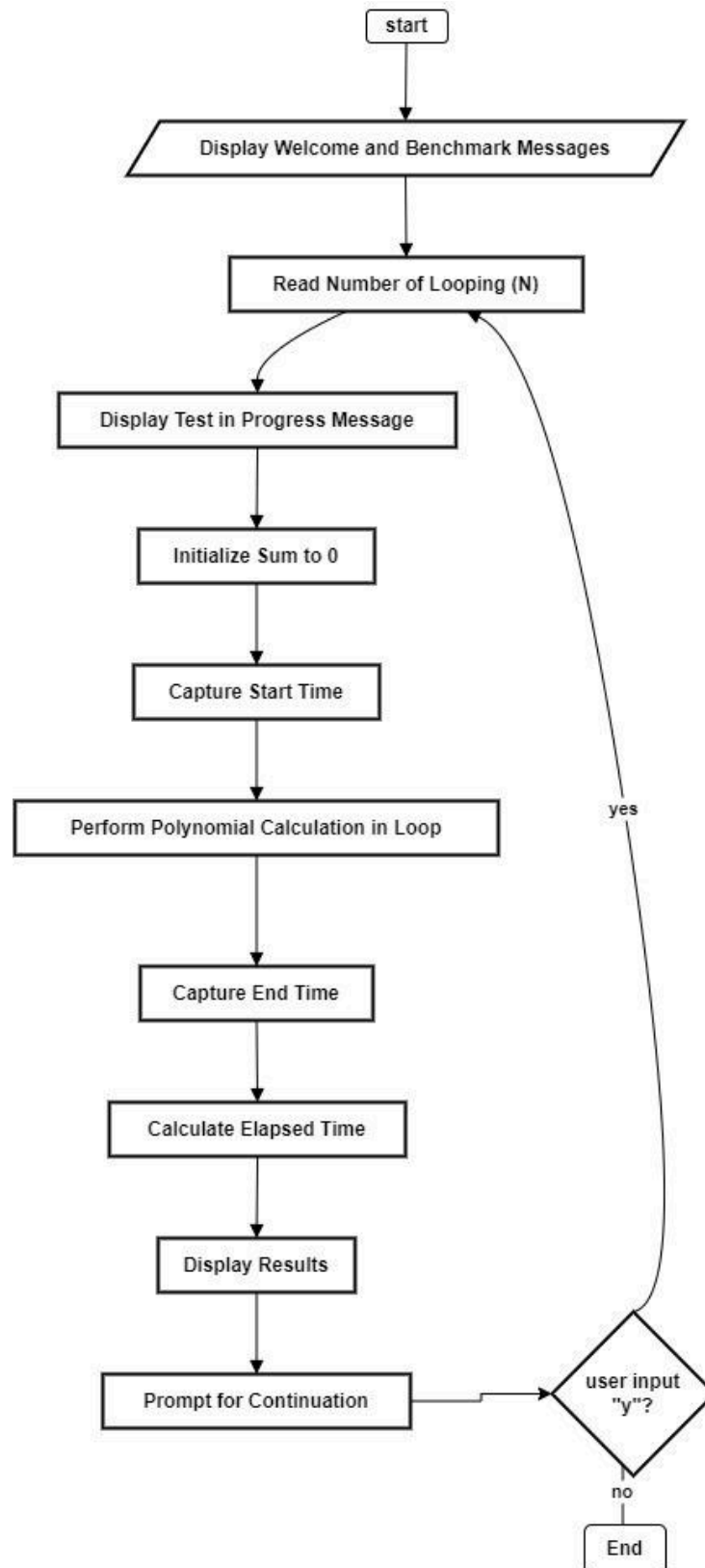
MiniTool Partition Wizard is used to execute disk benchmarking stress tests for Read and Write to the External Disk (HDD or SSD) of the computer. In this project, different mode tests which are Sequential test mode and Random test mode are used to evaluate the performance of storage devices and their influence on overall system efficiency.

2.0 Flowchart of the Project

2.1 CPU Benchmarking and Performance Analysis Using Free Tools



2.2 Writing ASM Code Program for Benchmarking Software



3.0 Benchmarking Results

3.1 CPU-Z

Lenovo IdeaPad Flex 5

CPU-Z			
CPU Mainboard Memory SPD Graphics Bench About			
Processor			
Name	Intel Core i5 1135G7		
Code Name	Tiger Lake-U	Max TDP	28.0 W
Package	Socket 1449 FCBGA		
Technology	10 nm	Core VID	0.692 V
Specification			
11th Gen Intel® Core™ i5-1135G7 @ 2.40GHz			
Family	6	Model	C
Ext. Family	6	Ext. Model	8C
Instructions	MMX, SSE, SSE2, SSE3, SSSE3, SSE4.1, SSE4.2, EM64T, AES, AVX, AVX2, AVX512, FMA3, SHA	Stepping	1
Clocks (Core #0)			
Core Speed	1197.20 MHz	Cache	
Multiplier	x 12.0 (4.0 - 42.0)	L1 Data	4 x 48 KBytes 12-way
Bus Speed	99.74 MHz	L1 Inst.	4 x 32 KBytes 8-way
Rated FSB		Level 2	4 x 1.25 MBytes 20-way
		Level 3	8 MBytes 8-way
Selection Socket #1 Cores 4 Threads 8			
CPU-Z Ver. 2.09.0.x64 Tools Validate Close			

CPU-Z

CPU | Mainboard | Memory | SPD | Graphics | Bench | About

Motherboard

ManufacturerLENOVO

ModelLNVNB161216SDK0J40700 WIN

Bus Specs.PCI-Express 4.0 (16.0 GT/s)

ChipsetIntelTiger LakeRev. 01

SouthbridgeIntelTiger Lake-U/Y PCHRev. 20

LPCIO

BIOS

BrandLENOVO

VersionFXCN31WW

Date07/30/2021

Graphic Interface

Bus

Transfer Rate

Max. Supported

Side Band Addressing

Max. Supported

CPU-Z Ver. 2.09.0.x64

Tools

Validate

Close

CPU-Z			
CPU Mainboard Memory SPD Graphics Bench About			
General			
Type	DDR4	Channel #	2 x 64-bit
Size	8 GBytes	Mem Controller Freq.	531.9 MHz
Timings			
DRAM Frequency	1063.7 MHz		
FSB:DRAM	3:32		
CAS# Latency (CL)	15.0 clocks		
RAS# to CAS# Delay (tRCD)	15 clocks		
RAS# Precharge (tRP)	15 clocks		
Cycle Time (tRAS)	35 clocks		
Row Refresh Cycle Time (tRFC)	374 clocks		
Command Rate (CR)	2T		
DRAM Idle Timer			
Total CAS# (tRDRAM)			
Row To Column (tRCD)			
CPU-Z Ver. 2.09.0.x64 Tools Validate Close			

ASUS VivoBook S14

CPU-Z			
CPU Mainboard Memory SPD Graphics Bench About			
Processor			
Name	Intel Core i5 8250U		
Code Name	Kaby Lake-R	Max TDP	15.0 W
Package	Socket 1356 FCBGA		
Technology	14 nm	Core VID	0.714 V
Specification			
Intel® Core™ i5-8250U CPU @ 1.60GHz			
Family	6	Model	E
Ext. Family	6	Ext. Model	8E
Instructions	MMX, SSE, SSE2, SSE3, SSSE3, SSE4.1, SSE4.2, EM64T, VT-x, AES, AVX2, FMA3	Stepping	A
Clocks (Core #0)			
Core Speed	897.80 MHz	Cache	
Multiplier	x 9.0 (4.0 - 34.0)	L1 Data	4 x 32 KBytes 8-way
Bus Speed	99.76 MHz	L1 Inst.	4 x 32 KBytes 8-way
Rated FSB		Level 2	4 x 256 KBytes 4-way
		Level 3	6 MBytes 12-way
Selection Socket #1 Cores 4 Threads 8			
CPU-Z Ver. 2.09.0.x64 Tools Validate Close			

CPU-Z										
CPU					Mainboard	Memory	SPD	Graphics	Bench	About
Motherboard										
Manufacturer	ASUSTeK COMPUTER INC.									
Model	X430UN							1.0		
Bus Specs.	PCI-Express 3.0 (8.0 GT/s)									
Chipset	Intel	Kaby Lake							Rev.	08
Southbridge	Intel	Coffee Lake-U/Y PCH							Rev.	21
LPCIO										
BIOS										
Brand	American Megatrends Inc.									
Version	X430UN.306									
Date	04/28/2020									
Graphic Interface										
Bus	PCI-Express									
Current Link Width	Max. Supported									
Side Band Addressing	Max. Supported									
CPU-Z Ver. 2.09.0.x64 Tools Validate Close										

CPU-Z			
CPU Mainboard Memory SPD Graphics Bench About			
General			
Type	DDR4	Channel #	Dual
Size	12 GBytes	DC Mode	
Timings			
DRAM Frequency	1197.7 MHz		
FSB:DRAM	3:36		
CAS# Latency (CL)	17.0 clocks		
RAS# to CAS# Delay (tRCD)	17 clocks		
RAS# Precharge (tRP)	17 clocks		
Cycle Time (tRAS)	39 clocks		
Row Refresh Cycle Time (tRFC)	420 clocks		
Command Rate (CR)	2T		
DRAM Idle Timer			
Total CAS# (tRDRAM)			
Row To Column (tRCD)			
CPU-Z Ver. 2.09.0.x64 Tools Validate Close			

Acer Nitro 5 Intel

CPU-Z			
CPU Mainboard Memory SPD Graphics Bench About			
Processor			
Name	Intel Core i7 11800H		
Code Name	Tiger Lake	Max TDP	45.0 W
Package	Socket 1787 FCBGA		
Technology	10 nm	Core VID	1.142 V
Specification			
11th Gen Intel® Core™ i7-11800H @ 2.30GHz			
Family	6	Model	D
Ext. Family	6	Ext. Model	8D
Instructions	MMX, SSE, SSE2, SSE3, SSSE3, SSE4.1, SSE4.2, EM64T, AES, AVX, AVX2, AVX512, FMA3, SHA	Stepping	1
Clocks (Core #0)			
Core Speed	4199.31 MHz	Cache	
Multiplier	x 42.0 (8.0 - 46.0)	L1 Data	8 x 48 KBytes 12-way
Bus Speed	99.98 MHz	L1 Inst.	8 x 32 KBytes 8-way
Rated FSB		Level 2	8 x 1.25 MBytes 20-way
		Level 3	24 MBytes 12-way
Selection Socket #1 Cores 8 Threads 16			
CPU-Z Ver. 2.09.0.x64 Tools Validate Close			

CPU-Z

CPU | Mainboard | Memory | SPD | Graphics | Bench | About

Motherboard

Manufacturer: TGL

Model: Scale_TLS V1.17

Bus Specs. PCI-Express 4.0 (16.0 GT/s)

Chipset: Intel Tiger Lake-H Rev. 05

Southbridge: Intel HM570 Rev. 11

LPCIO:

BIOS

Brand: Insyde Corp.

Version: V1.17

Date: 06/02/2022

Graphic Interface

Bus: PCI-Express

Current Link Width: Max. Supported

Side Band Addressing: Max. Supported

CPU-Z Ver. 2.09.0.x64 Tools Validate Close

CPU-Z			
CPU Mainboard Memory SPD Graphics Bench About			
General			
Type	DDR4	Channel #	Single
Size	16 GBytes	DC Mode	
Timings			
DRAM Frequency			
FSB:DRAM	3:48		
CAS# Latency (CL)	22.0 clocks		
RAS# to CAS# Delay (tRCD)	22 clocks		
RAS# Precharge (tRP)	22 clocks		
Cycle Time (tRAS)	52 clocks		
Row Refresh Cycle Time (tRFC)	560 clocks		
Command Rate (CR)	1T		
DRAM Idle Timer			
Total CAS# (tRDRAM)			
Row To Column (tRCD)			
CPU-Z Ver. 2.09.0.x64 Tools Validate Close			

Acer Nitro AN515-55

CPU-Z

CPU

Mainboard

Memory

SPD

Graphics

Bench

About

Processor

Name

Intel Core i7 10750H

Code Name

Comet Lake

Max TDP

45.0 W

Package

Socket 1440 FCBGA

Technology

14 nm

Core VID

0.672 V

Specification

Intel® Core™ i7-10750H CPU @ 2.60GHz

Family

6

Model

5

Stepping

2

Ext. Family

6

Ext. Model

A5

Revision

R0/R1

Instructions

MMX, SSE, SSE2, SSE3, SSSE3, SSE4.1, SSE4.2, EM64T, VT-x, AES, AVX, AVX2, FMA3

Clocks (Core #0)

Core Speed

4489.02 MHz

Multiplier

x45.0 (8.0 - 48.0)

Bus Speed

99.76 MHz

Rated FSB

Cache

L1 Data

6 x 32 KBytes

8-way

L1 Inst.

6 x 32 KBytes

8-way

Level 2

6 x 256 KBytes

4-way

Level 3

12 MBytes

16-way

Selection

Socket #1

Cores

6

Threads

12

CPU-Z

Ver. 2.09.0.x64

Tools

Validate

Close

CPU-Z

CPU

Mainboard

Memory

SPD

Graphics

Bench

About

Motherboard

Manufacturer

CML

Model

Stonic_CMS

V2.06

Bus Specs.

PCI-Express 3.0 (8.0 GT/s)

Chipset

Intel

Comet Lake

Rev.

02

Southbridge

Intel

HM470

Rev.

00

LPCIO

BIOS

Brand

Insyde Corp.

Version

V2.06

Date

08/19/2021

Graphic Interface

Bus

PCI-Express 3.0

Current Link Width

x16

Max. Supported

x16

Current Link Speed

2.5 GT/s

Max. Supported

8.0 GT/s

CPU-Z

Ver. 2.09.0.x64

Tools

Validate

Close

CPU-Z

CPU

Mainboard

Memory

SPD

Graphics

Bench

About

General

Type

DDR4

Channel #

Dual

Size

16 GBytes

DC Mode

Uncore Frequency

2793.8 MHz

Timings

DRAM Frequency

1463.4 MHz

FSB:DRAM

3:44

CAS# Latency (CL)

21.0 clocks

RAS# to CAS# Delay (TRCD)

21 clocks

RAS# Precharge (TRP)

21 clocks

Cycle Time (TRAS)

47 clocks

Row Refresh Cycle Time (tRFC)

514 clocks

Command Rate (CR)

2T

DRAM Idle Timer

Total CAS# (tRDRAM)

Row To Column (tRCD)

CPU-Z

Ver. 2.09.0.x64

Tools

Validate

Close

3.2 Geekbench

Lenovo IdeaPad Flex 5

System Information

System Information	
Operating System	Microsoft Windows 11 Home (64-bit)
Model	LENOVO 82HS
Motherboard	LENOVO LNVNB161216
Power Plan	Balanced

CPU Information

Name	Intel Core i5-1135G7
Topology	1 Processor, 4 Cores, 8 Threads
Identifier	GenuineIntel Family 6 Model 140 Stepping 1
Base Frequency	2.42 GHz
Cluster 1	4 Cores
Maximum Frequency	4190 MHz
Package	Socket 1449 FCBGA
Codename	Tiger Lake-U
L1 Instruction Cache	32.0 KB x 4
L1 Data Cache	48.0 KB x 4
L2 Cache	1.25 MB x 4
L3 Cache	8.00 MB x 1

Memory Information

Size	7.56 GB
Transfer Rate	2128 MT/s
Type	DDR4 SDRAM
Channels	2

Single-Core Performance

Single-Core Score	376
File Compression	361 51.8 MB/sec
Navigation	325 1.96 routes/sec
HTML5 Browser	459 9.39 pages/sec
PDF Renderer	403 9.29 Mpixels/sec
Photo Library	351 4.77 images/sec
Clang	358 1.76 Klines/sec
Text Processing	332 26.6 pages/sec
Asset Compression	368 11.4 MB/sec
Object Detection	350 10.5 images/sec
Background Blur	457 1.89 images/sec
Horizon Detection	521 16.2 Mpixels/sec
Object Remover	316 24.3 Mpixels/sec
HDR	374 11.0 Mpixels/sec
Photo Filter	439 4.36 images/sec
Ray Tracer	325 314.0 Kpixels/sec
Structure from Motion	372 11.8 Kpixels/sec

Multi-Core Performance

Multi-Core Score	843
File Compression	648 93.1 MB/sec
Navigation	923 5.56 routes/sec
HTML5 Browser	925 18.9 pages/sec
PDF Renderer	896 20.7 Mpixels/sec
Photo Library	994 13.5 images/sec
Clang	911 4.49 Klines/sec
Text Processing	366 29.3 pages/sec
Asset Compression	1109 34.4 MB/sec
Object Detection	642 19.2 images/sec
Background Blur	854 3.54 images/sec
Horizon Detection	1396 43.5 Mpixels/sec
Object Remover	729 56.0 Mpixels/sec
HDR	930 27.3 Mpixels/sec
Photo Filter	967 9.60 images/sec
Ray Tracer	947 916.6 Kpixels/sec
Structure from Motion	970 30.7 Kpixels/sec

Asus Vivobook S14

System Information

System Information	
Operating System	Microsoft Windows 11 Home Single Language (64-bit)
Model	ASUSTeK COMPUTER INC. VivoBook S14 X430UN
Motherboard	ASUSTeK COMPUTER INC. X430UN
Power Plan	Balanced

CPU Information	
Name	Intel Core i5-8250U
Topology	1 Processor, 4 Cores, 8 Threads
Identifier	GenuineIntel Family 6 Model 142 Stepping 10
Base Frequency	1.80 GHz
Cluster 1	4 Cores
Maximum Frequency	3392 MHz
Package	Socket 1356 FCBGA
Codename	Kaby Lake-R
L1 Instruction Cache	32.0 KB x 4
L1 Data Cache	32.0 KB x 4
L2 Cache	256 KB x 4
L3 Cache	6.00 MB x 1

Memory Information	
Size	11.88 GB
Transfer Rate	1330 MT/s
Type	DDR4 SDRAM
Channels	2

Single-Core Performance	Multi-Core Performance
Single-Core Score1201	Multi-Core Score2994
File Compression1119160.8 MB/sec	File Compression1430205.4 MB/sec
Navigation13798.31 routes/sec	Navigation437726.4 routes/sec
HTML5 Browser133527.3 pages/sec	HTML5 Browser344770.6 pages/sec
PDF Renderer132830.6 Mpixels/sec	PDF Renderer415795.9 Mpixels/sec
Photo Library100713.7 images/sec	Photo Library299640.7 images/sec
Clang12676.24 Klines/sec	Clang421420.8 Klines/sec
Text Processing111389.2 pages/sec	Text Processing1416113.4 pages/sec
Asset Compression133141.2 MB/sec	Asset Compression4634143.6 MB/sec
Object Detection54216.2 images/sec	Object Detection130038.9 images/sec
Background Blur16756.93 images/sec	Background Blur379315.7 images/sec
Horizon Detection173454.0 Mpixels/sec	Horizon Detection4134128.6 Mpixels/sec
Object Remover103679.7 Mpixels/sec	Object Remover2862220.0 Mpixels/sec
HDR124836.6 Mpixels/sec	HDR329396.6 Mpixels/sec
Photo Filter161016.0 images/sec	Photo Filter295629.3 images/sec
Ray Tracer10521.02 Mpixels/sec	Ray Tracer41233.99 Mpixels/sec
Structure from Motion137843.6 Kpixels/sec	Structure from Motion3264103.4 Kpixels/sec

Acer Nitro 5 Intel

System Information	
Operating System	Microsoft Windows 11 Home (64-bit)
Model	Acer Nitro AN515-57
Motherboard	TGL Scala_TLS
Power Plan	High performance
CPU Information	
Name	Intel Core i7-11800H
Topology	1 Processor, 8 Cores, 16 Threads
Identifier	GenuineIntel Family 6 Model 141 Stepping 1
Base Frequency	2.30 GHz
Cluster 1	8 Cores
Maximum Frequency	4589 MHz
Package	Socket 1787 FCBGA
Codename	Tiger Lake
L1 Instruction Cache	32.0 KB x 8
L1 Data Cache	48.0 KB x 8
L2 Cache	1.25 MB x 8
L3 Cache	24.0 MB x 1
Memory Information	
Size	16.00 GB
Transfer Rate	3190 MT/s
Type	DDR4 SDRAM
Channels	1

Single-Core Performance		Multi-Core Performance	
Single-Core Score	1206	Multi-Core Score	4282
File Compression	1249 179.3 MB/sec	File Compression	2088 299.9 MB/sec
Navigation	1160 6.99 routes/sec	Navigation	6073 36.6 routes/sec
HTML5 Browser	1226 25.1 pages/sec	HTML5 Browser	3785 77.5 pages/sec
PDF Renderer	1175 27.1 Mpixels/sec	PDF Renderer	5701 131.5 Mpixels/sec
Photo Library	1137 15.4 images/sec	Photo Library	5536 75.1 images/sec
Clang	1170 5.76 Klines/sec	Clang	6685 32.9 Klines/sec
Text Processing	1036 83.0 pages/sec	Text Processing	1394 111.6 pages/sec
Asset Compression	1152 35.7 MB/sec	Asset Compression	7183 222.6 MB/sec
Object Detection	1176 35.2 images/sec	Object Detection	2472 74.0 images/sec
Background Blur	1387 5.74 images/sec	Background Blur	4259 17.6 images/sec
Horizon Detection	1647 51.2 Mpixels/sec	Horizon Detection	5929 184.5 Mpixels/sec
Object Remover	1042 80.1 Mpixels/sec	Object Remover	4853 373.2 Mpixels/sec
HDR	1200 35.2 Mpixels/sec	HDR	4523 132.7 Mpixels/sec
Photo Filter	1367 13.6 images/sec	Photo Filter	3415 33.9 images/sec
Ray Tracer	1171 1.13 Mpixels/sec	Ray Tracer	7802 7.55 Mpixels/sec
Structure from Motion	1305 41.3 Kpixels/sec	Structure from Motion	4618 146.2 Kpixels/sec

Acer Nitro AN515-55

Result Information	
User	kinan
Upload Date	July 01 2024 09:10 AM
Views	1
System Information	
System Information	
Operating System	Microsoft Windows 11 Pro (64-bit)
Model	Acer Nitro AN515-55
Motherboard	CML Stonic_CMS
Power Plan	Power saver
CPU Information	
Name	Intel Core i7-10750H
Topology	1 Processor, 6 Cores, 12 Threads
Identifier	GenuineIntel Family 6 Model 165 Stepping 2
Base Frequency	2.59 GHz
Cluster 1	6 Cores
Maximum Frequency	4788 MHz
Package	Socket 1440 FCBGA
Codename	Comet Lake
L1 Instruction Cache	32.0 KB x 6
L1 Data Cache	32.0 KB x 6
L2 Cache	256 KB x 6
L3 Cache	12.0 MB x 1
Memory Information	
Size	16.00 GB
Type	DDR4 SDRAM
Channels	2

Single-Core Performance

Single-Core Score	1254
File Compression	1266 181.9 MB/sec
Navigation	1396 8.41 routes/sec
HTML5 Browser	1382 28.3 pages/sec
PDF Renderer	1349 31.1 Mpixels/sec
Photo Library	1047 14.2 images/sec
Clang	1336 6.58 Klines/sec
Text Processing	1240 99.3 pages/sec
Asset Compression	1403 43.5 MB/sec
Object Detection	558 16.7 images/sec
Background Blur	1744 7.22 images/sec
Horizon Detection	1827 56.9 Mpixels/sec
Object Remover	1112 85.5 Mpixels/sec
HDR	1290 37.9 Mpixels/sec
Photo Filter	1645 16.3 images/sec
Ray Tracer	1021 987.5 Kpixels/sec
Structure from Motion	1384 43.8 Kpixels/sec

Multi-Core Performance

Multi-Core Score	5465
File Compression	3104 445.8 MB/sec
Navigation	6822 41.1 routes/sec
HTML5 Browser	6340 129.8 pages/sec
PDF Renderer	6752 155.7 Mpixels/sec
Photo Library	6118 83.0 images/sec
Clang	8590 42.3 Klines/sec
Text Processing	1593 127.6 pages/sec
Asset Compression	9468 293.4 MB/sec
Object Detection	2316 69.3 images/sec
Background Blur	7013 29.0 images/sec
Horizon Detection	8277 257.6 Mpixels/sec
Object Remover	5649 434.3 Mpixels/sec
HDR	5887 172.8 Mpixels/sec
Photo Filter	4920 48.8 images/sec
Ray Tracer	9094 8.80 Mpixels/sec
Structure from Motion	6744 213.5 Kpixels/sec

3.3 Task Manager

Lenovo IdeaPad Flex 5

Low Load (Open Word Editing)

Task Manager		Type a name, publisher, or PID to search			
Processes					
		14%	60%	13%	0%
Name	Status	CPU	Memory	Disk	Network
Apps (2)					
> Microsoft Word		3.6%	100.5 MB	6.6 MB/s	0.1 Mbps
> Task Manager		0%	66.8 MB	0.1 MB/s	0 Mbps
Background processes (84)					
> Acrobat Update Service (32 bit)		0%	1.1 MB	0 MB/s	0 Mbps
> AgentService		0%	0.8 MB	0 MB/s	0 Mbps
> Antimalware Core Service		0%	5.3 MB	0 MB/s	0 Mbps
> Antimalware Service Executable		0%	187.7 MB	0.1 MB/s	0 Mbps
> Apache HTTP Server (2)		0%	2.2 MB	0 MB/s	0 Mbps
Application Frame Host		0%	7.7 MB	0 MB/s	0 Mbps
Artificial Intelligence (AI) Host ...		0%	19.7 MB	0 MB/s	0 Mbps
COM Surrogate		0%	2.0 MB	0 MB/s	0 Mbps
COM Surrogate		0%	3.4 MB	0 MB/s	0 Mbps
CTF Loader		0%	4.3 MB	0 MB/s	0 Mbps
> DAX API (2)		0%	5.2 MB	0 MB/s	0 Mbps
Device Association Framework...		0%	11.9 MB	0 MB/s	0 Mbps
Device Association Framework...		0%	1.3 MB	0 MB/s	0 Mbps
> Fortemedia Service		0%	1.9 MB	0 MB/s	0 Mbps

Middle Load

<

High Load (Rendering a Video)

ASUS VivoBook S14

Low CPU load (downloading files)

<

Mixed CPU load (playing a game while watching youtube)

High CPU load (Rendering a Video)

Acer Nitro 5 Intel
Low Load (Browsing Documents)

Task Manager						
Type a name, publisher, or Pl...						
Processes						
Run new task End task Efficiency mode						
		13%	58%	0%	0%	
Name		CPU	Memory	Disk	Network	
Apps (2)						
Task Manager		0.2%	87.5 MB	0 MB/s	0 Mbps	
Windows Explorer		0%	251.6 MB	0 MB/s	0 Mbps	
Background processes (133)						
µTorrent Helper (32 bit)		0%	0.6 MB	0 MB/s	0 Mbps	
µTorrent Web (32 bit)		0%	3.4 MB	0 MB/s	0 Mbps	
ACCSvc		0%	0.3 MB	0 MB/s	0 Mbps	
Activation Licensing Service		0%	1.5 MB	0 MB/s	0 Mbps	
Agent for EasyConnect (32 bit)		0%	3.2 MB	0 MB/s	0 Mbps	
AgentService.exe		0%	3.7 MB	0 MB/s	0 Mbps	
Antimalware Core Service		0%	6.7 MB	0 MB/s	0 Mbps	
Antimalware Service Executable		0.1%	194.1 MB	0.1 MB/s	0 Mbps	
Application Frame Host		0%	2.2 MB	0 MB/s	0 Mbps	
AppMonitorPlugIn		0%	8.0 MB	0 MB/s	0 Mbps	
Bonjour Service		0%	1.5 MB	0 MB/s	0 Mbps	
CamUsage		0%	2.0 MB	0 MB/s	0 Mbps	
Canon IJ Network Scanner Sel...		0%	0.2 MB	0 MB/s	0 Mbps	
Casting protocol connection li...		0%	1.9 MB	0 MB/s	0 Mbps	
CDA Server		0%	0.2 MB	0 MB/s	0 Mbps	
COM Surrogate		0%	0.2 MB	0 MB/s	0 Mbps	
COM Surrogate		0%	0.1 MB	0 MB/s	0 Mbps	
COM Surrogate		0%	3.0 MB	0 MB/s	0 Mbps	
COM Surrogate		0%	4.1 MB	0 MB/s	0 Mbps	

Middle load

Task Manager						
Type a name, publisher, or P...						
Processes						
Run new task End task Efficiency mode						
		19%	65%	3%	0%	
		CPU	Memory	Disk	Network	
Apps (2)						
> Opera GX Internet Browser (19)		0.3%	1,264.8 MB	0.1 MB/s	0.1 Mbps	
> Task Manager		0.6%	110.1 MB	0.1 MB/s	0 Mbps	
Background processes (134)						
µTorrent Helper (32 bit)		0%	1.7 MB	0 MB/s	0 Mbps	
µTorrent Web (32 bit)		0%	5.1 MB	0 MB/s	0 Mbps	
> ACCSvc		0%	0.4 MB	0 MB/s	0 Mbps	
> Acer Purified Voice Console (R...		0%	0.6 MB	0 MB/s	0 Mbps	
> Activation Licensing Service		0%	1.7 MB	0 MB/s	0 Mbps	
Agent for EasyConnect (32 bit)		0%	3.9 MB	0 MB/s	0 Mbps	
> AgentService.exe		0%	0.5 MB	0 MB/s	0 Mbps	
> Antimalware Core Service		0%	9.5 MB	0 MB/s	0 Mbps	
> Antimalware Service Executable		0.1%	275.1 MB	0.1 MB/s	0 Mbps	
Application Frame Host		0%	3.3 MB	0 MB/s	0 Mbps	
AppMonitorPlugIn		0%	7.6 MB	0 MB/s	0 Mbps	
> Bonjour Service		0%	2.2 MB	0 MB/s	0 Mbps	
CamUsage		0%	2.4 MB	0 MB/s	0 Mbps	

High Load (>70% Memory)

Task Manager						
Type a name, publisher, or PI...						
Processes						
Run new task End task Efficiency mode						
		14%	73%	0%	0%	
		CPU	Memory	Disk	Network	
Name		Status				
Apps (3)						
>	Opera GX Internet Browser (29)		1.0%	2,208.8 MB	0.2 MB/s	0.1 Mbps
>	Task Manager		3.0%	87.7 MB	0 MB/s	0 Mbps
>	Windows Explorer (2)		0.1%	240.5 MB	0 MB/s	0 Mbps
Background processes (134)						
	µTorrent Helper (32 bit)		0%	0.6 MB	0 MB/s	0 Mbps
	µTorrent Web (32 bit)		0%	3.4 MB	0 MB/s	0.1 Mbps
>	ACCSvc		0%	0.3 MB	0 MB/s	0 Mbps
>	Activation Licensing Service		0%	1.5 MB	0 MB/s	0 Mbps
	Agent for EasyConnect (32 bit)		0%	3.2 MB	0 MB/s	0 Mbps
>	AgentService.exe		0%	3.7 MB	0 MB/s	0 Mbps
>	Antimalware Core Service		0%	6.7 MB	0 MB/s	0 Mbps
>	Antimalware Service Executable		0%	197.3 MB	0.1 MB/s	0 Mbps
	Application Frame Host		0%	2.2 MB	0 MB/s	0 Mbps
	AppMonitorPlugIn		0%	7.9 MB	0 MB/s	0 Mbps
>	Bonjour Service		0%	1.5 MB	0 MB/s	0 Mbps
	CamUsage		0%	2.0 MB	0 MB/s	0 Mbps
	Canon IJ Network Scanner Sel...		0%	0.2 MB	0 MB/s	0 Mbps
	Casting protocol connection li...		0%	1.9 MB	0 MB/s	0 Mbps
	CDA Server		0%	0.2 MB	0 MB/s	0 Mbps
	COM Surrogate		0%	0.2 MB	0 MB/s	0 Mbps
	COM Surrogate		0%	0.1 MB	0 MB/s	0 Mbps
	COM Surrogate		0%	2.9 MB	0 MB/s	0 Mbps

Acer Nitro AN515-55

Low load

Type a name, publisher, or PID to search					
Processes					
Name	Status	5% CPU	48% Memory	1% Disk	0% Network
Apps (4)					
> Brave Browser (21)		1.1%	1,512.4 MB	0.1 MB/s	0.1 Mbps
> Microsoft Word		0%	90.6 MB	0 MB/s	0 Mbps
> SnippingTool.exe		0%	68.7 MB	0 MB/s	0 Mbps
> Task Manager		2.3%	81.0 MB	0 MB/s	0 Mbps
Background processes (86)					
> Adobe Acrobat Update Service (...)		0%	0.4 MB	0 MB/s	0 Mbps
> AgentService.exe		0%	1.5 MB	0 MB/s	0 Mbps
> Antimalware Core Service		0%	8.2 MB	0 MB/s	0.1 Mbps
> Antimalware Service Executable		0.1%	158.9 MB	0.1 MB/s	0 Mbps
> AnyDesk (32 bit)		0%	5.1 MB	0 MB/s	0 Mbps
Application Frame Host		0%	5.6 MB	0 MB/s	0 Mbps
Artificial Intelligence (AI) Host for ...		0%	19.7 MB	0 MB/s	0 Mbps
BraveSoftware Update		0%	0.5 MB	0 MB/s	0 Mbps
BraveSoftware Update (32 bit)		0%	0.6 MB	0 MB/s	0 Mbps
BraveSoftware Update (32 bit)		0%	0.8 MB	0 MB/s	0 Mbps
Cisco Webex Meetings (32 bit)		0%	2.9 MB	0 MB/s	0 Mbps
COM Surrogate		0%	0.9 MB	0 MB/s	0 Mbps
COM Surrogate		0%	3.1 MB	0 MB/s	0 Mbps
COM Surrogate		0%	3.3 MB	0 MB/s	0 Mbps
> Controls the DTS audio processin...		0%	1.5 MB	0 MB/s	0 Mbps
> Creative Cloud Update Service (3...		0%	0.5 MB	0 MB/s	0 Mbps
CTF Loader		0%	4.8 MB	0 MB/s	0 Mbps
Device Association Framework Pr...		0%	3.4 MB	0 MB/s	0 Mbps

Normal load

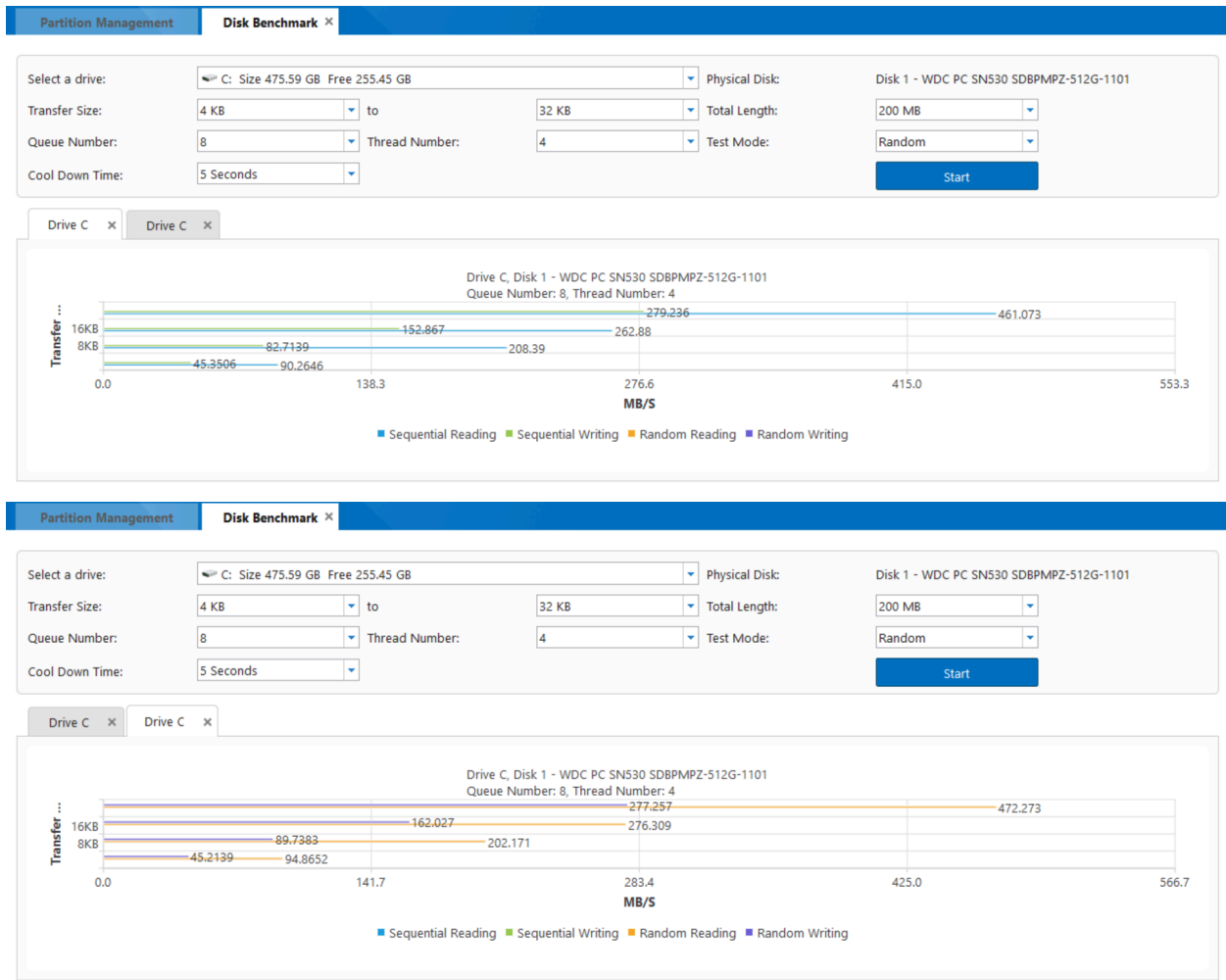
Task Manager						
Processes						
Name	Status	8% CPU	47% Memory	1% Disk	0% Network	
Apps (6)						
> Brave Browser (15)		0.8%	1,196.9 MB	0.1 MB/s	0 Mbps	
> Burp Suite Community Edition		0.6%	258.2 MB	0.8 MB/s	6.1 Mbps	
> Microsoft Word		0%	62.0 MB	0 MB/s	0 Mbps	
> SnippingTool.exe		1.1%	22.3 MB	0.1 MB/s	0 Mbps	
> Task Manager		1.4%	72.3 MB	0.1 MB/s	0 Mbps	
> Telegram Desktop (32 bit)		0%	167.7 MB	0 MB/s	0.1 Mbps	
Background processes (82)						
> Adobe Acrobat Update Service (...)		0%	0.7 MB	0 MB/s	0 Mbps	
> AgentService.exe		0%	4.3 MB	0 MB/s	0 Mbps	
> Antimalware Core Service		0%	6.4 MB	0 MB/s	0 Mbps	
> Antimalware Service Executable		1.0%	229.6 MB	0.1 MB/s	0 Mbps	
> AnyDesk (32 bit)		0%	26.2 MB	0 MB/s	0 Mbps	
Application Frame Host		0%	4.4 MB	0 MB/s	0 Mbps	
Artificial Intelligence (AI) Host for ...		0%	18.9 MB	0 MB/s	0 Mbps	
BraveSoftware Update		0%	0.5 MB	0 MB/s	0 Mbps	
BraveSoftware Update (32 bit)		0%	0.6 MB	0 MB/s	0 Mbps	
BraveSoftware Update (32 bit)		0%	0.8 MB	0 MB/s	0 Mbps	
Cisco Webex Meetings (32 bit)		0%	2.4 MB	0 MB/s	0 Mbps	
COM Surrogate		0%	2.6 MB	0 MB/s	0 Mbps	
> Controls the DTS audio processin...		0%	9.8 MB	0 MB/s	0 Mbps	
> Creative Cloud Update Service (3...		0%	0.7 MB	0 MB/s	0 Mbps	
CTF Loader		0%	4.5 MB	0 MB/s	0 Mbps	
Device Association Framework Pr...		0%	1.2 MB	0 MB/s	0 Mbps	
Device Association Framework Pr...		0%	4.1 MB	0 MB/s	0 Mbps	

High

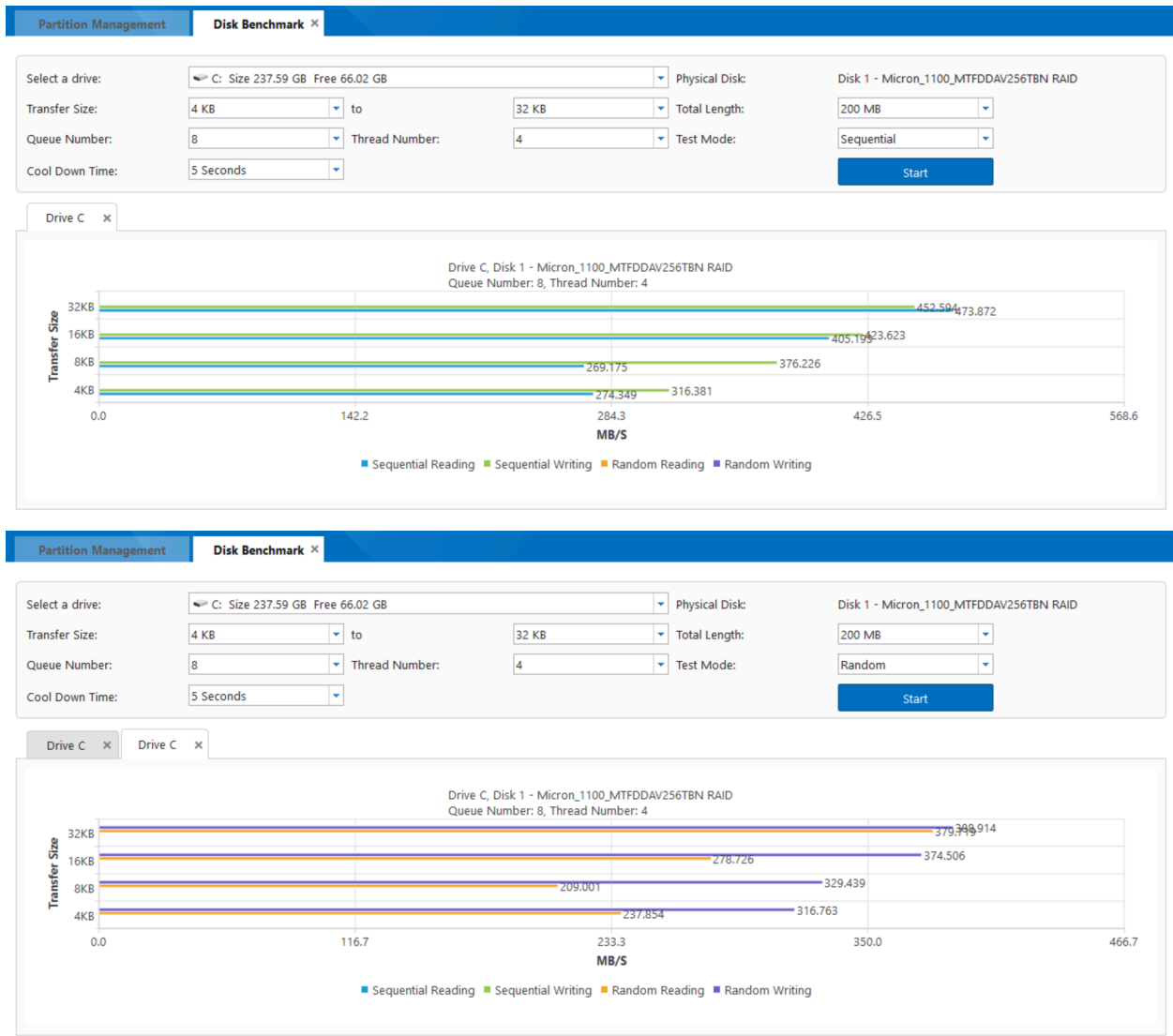
Task Manager						🔍 Type a name, published
Processes						
Name	Status	40% CPU	75% Memory	12% Disk	0% Network	
Apps (13)						
> Brave Browser (16)		1.1%	1,409.6 MB	0.4 MB/s	0.1 Mbps	
> Burp Suite Community Edition		0%	258.8 MB	0 MB/s	0 Mbps	
> Firefox (12)		1.1%	603.0 MB	0 MB/s	0.1 Mbps	
> Google Chrome (12)		0%	437.7 MB	0 MB/s	0 Mbps	
> Microsoft Edge (15)		1.3%	621.5 MB	0.5 MB/s	0.1 Mbps	
> Microsoft Word		0%	61.6 MB	0.1 MB/s	0.1 Mbps	
> Netflix (2)		0%	5.5 MB	0 MB/s	0 Mbps	
> Task Manager		0.6%	81.3 MB	0 MB/s	0 Mbps	
> Telegram Desktop (32 bit)		0%	167.6 MB	0 MB/s	0 Mbps	
> Visual Studio Code (12)		0%	613.8 MB	0 MB/s	0.1 Mbps	
> VMware Player (32 bit)		0%	19.5 MB	0 MB/s	0 Mbps	
> WhatsApp (2)		0%	95.6 MB	0 MB/s	0 Mbps	
> Windows Explorer (2)		0%	158.5 MB	0 MB/s	0 Mbps	
Background processes (96)						
> Adobe Acrobat Update Service (...)		0%	0.7 MB	0 MB/s	0 Mbps	
Adobe Crash Processor		0%	2.8 MB	0 MB/s	0.1 Mbps	
> AgentService.exe		0%	4.3 MB	0 MB/s	0 Mbps	
> Antimalware Core Service		0%	6.5 MB	0 MB/s	0 Mbps	
> Antimalware Service Executable		0.4%	271.9 MB	0.1 MB/s	0 Mbps	
> AnyDesk (32 bit)		0%	26.2 MB	0 MB/s	0 Mbps	
Application Frame Host		0%	13.9 MB	0 MB/s	0 Mbps	
Artificial Intelligence (AI) Host for ...		0%	18.9 MB	0 MB/s	0 Mbps	
BraveSoftware Update		0%	0.5 MB	0 MB/s	0 Mbps	
BraveSoftware Update (32 bit)		0%	0.6 MB	0 MB/s	0 Mbps	

3.4 MiniTools

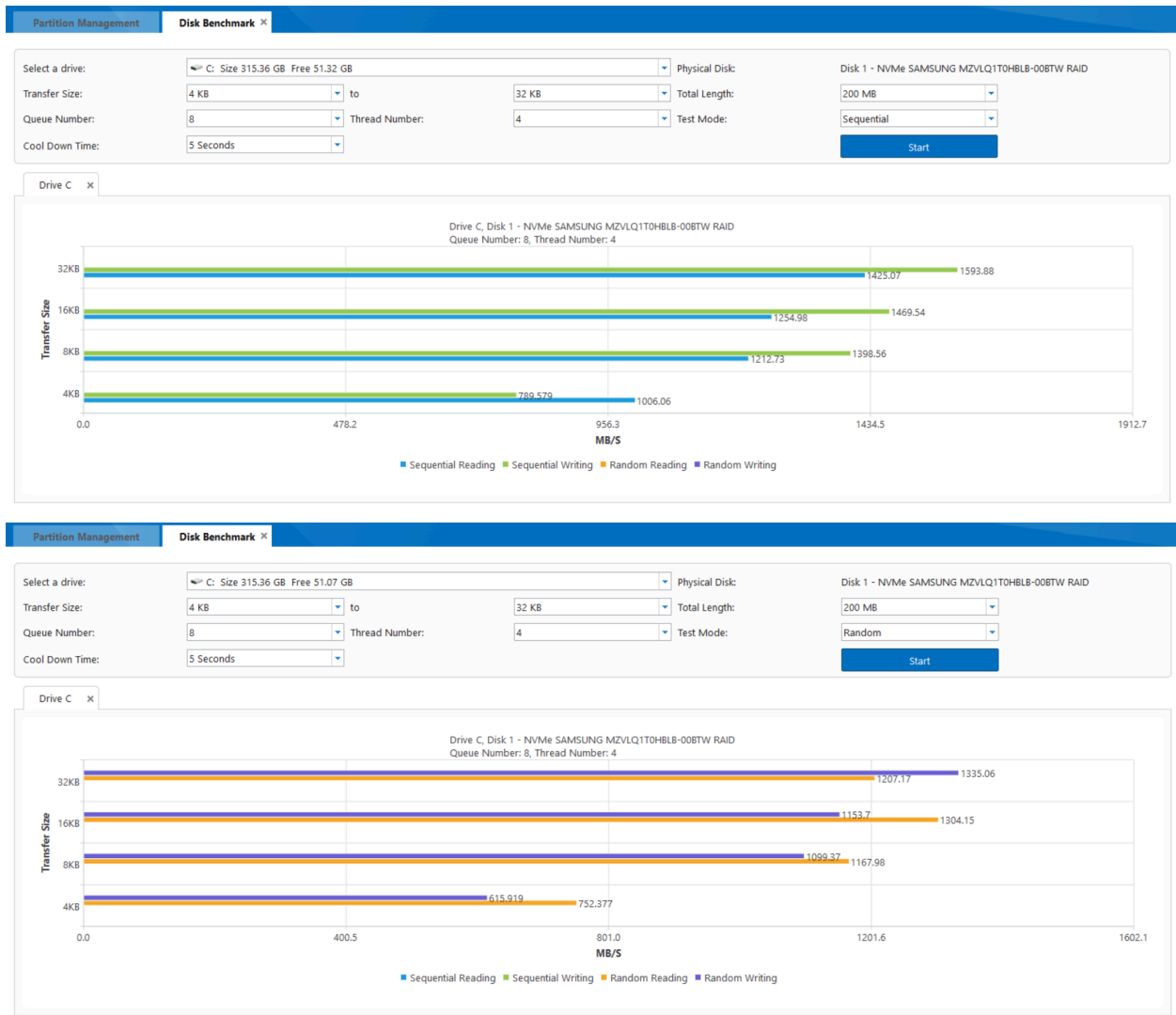
Lenovo IdeaPad Flex 5



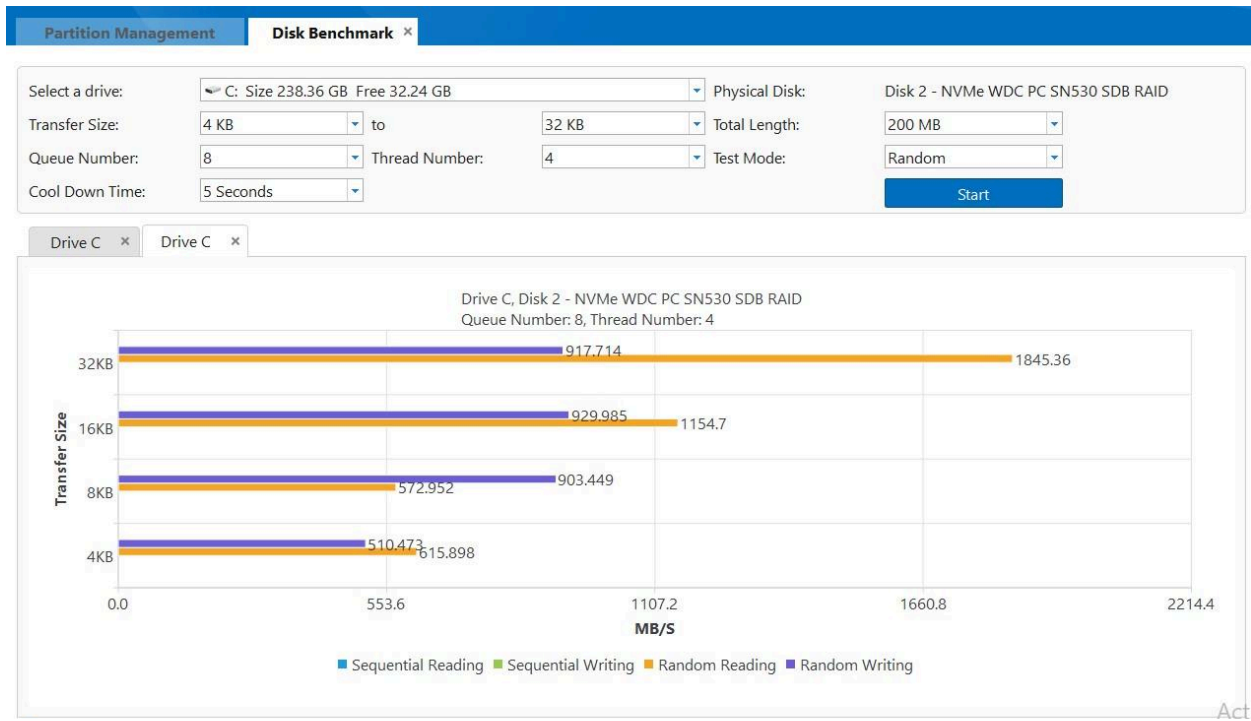
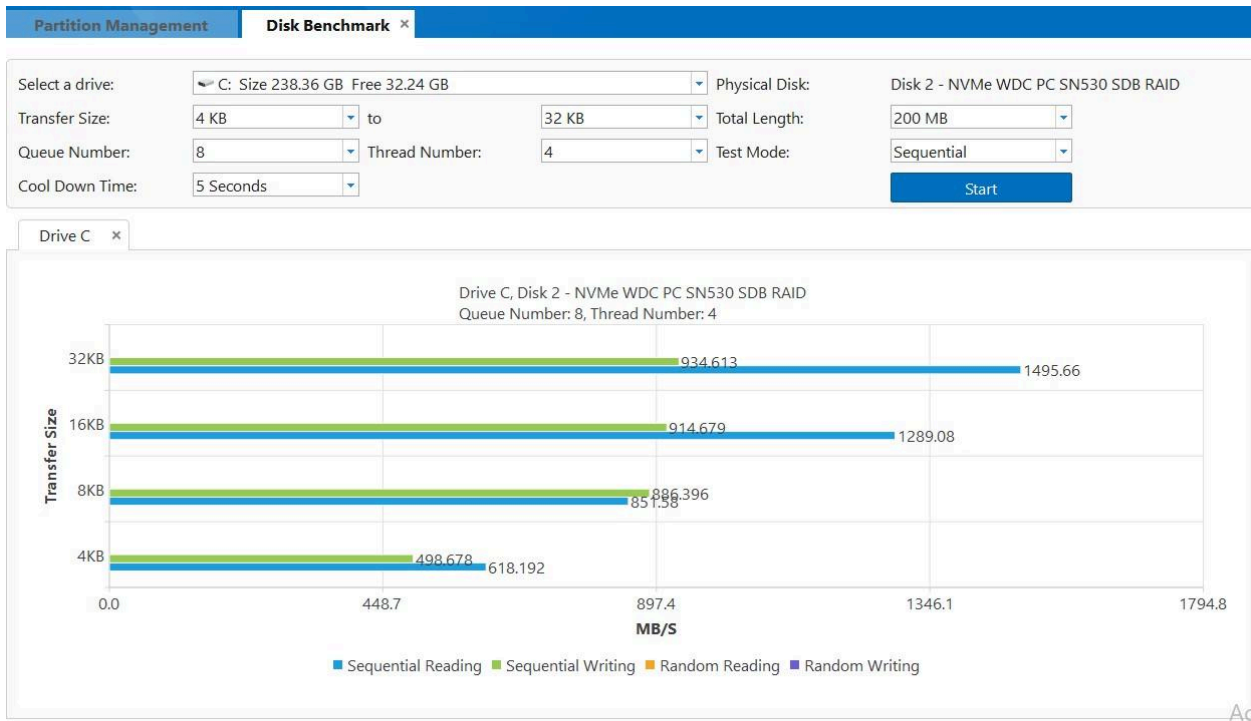
ASUS VivoBook S14



Acer Nitro 5 Intel



Acer Nitro AN515-55



3.5 ASM Code Program

Lenovo IdeaPad Flex 5

```
Welcome to CPU Benchmark Program
Benchmark CPU time Using Equation  $y = 7x^3 + 9x^2 + 13x + 25$ 
(with delay coef1, coef2, coef3, coef4 = 7, 9, 13, 25 msec)
Enter Number of Looping (N) = 50

CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 78089638
Second Capture Execution time in millisecond: 78095349
Different Execution time in millisecond: 5711
Value of Sum from the Stress Test (polynomial) = 11783525
```

```
CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 78105321
Second Capture Execution time in millisecond: 78113226
Different Execution time in millisecond: 7905
Value of Sum from the Stress Test (polynomial) = 181630800
Press 'y' to continue or 'n' to exit the benchmark:
```

```
Enter Number of Looping (N) = 200

CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 78124868
Second Capture Execution time in millisecond: 78140362
Different Execution time in millisecond: 15494
Value of Sum from the Stress Test (polynomial) = 2852516600
Press 'y' to continue or 'n' to exit the benchmark:
```

```
Enter Number of Looping (N) = 500

CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 78151152
Second Capture Execution time in millisecond: 78189989
Different Execution time in millisecond: 38837
Value of Sum from the Stress Test (polynomial) = 2816521600
Press 'y' to continue or 'n' to exit the benchmark:
```

```
Enter Number of Looping (N) = 700

CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 78198561
Second Capture Execution time in millisecond: 78252989
Different Execution time in millisecond: 54428
Value of Sum from the Stress Test (polynomial) = 1503975592
Press 'y' to continue or 'n' to exit the benchmark:
```

```
Enter Number of Looping (N) = 1000

CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 78259387
Second Capture Execution time in millisecond: 78337021
Different Execution time in millisecond: 77634
Value of Sum from the Stress Test (polynomial) = 4166126232
```

ASUS VivoBook S14

```
C:\Users\Admin\source\repos x + v
Welcome to CPU Benchmark Program
Benchmark CPU time Using Equation  $y = 7x^3 + 9x^2 + 13x + 25$ 
(with delay coef1, coef2, coef3, coef4 = 7, 9, 13, 25 msec)
Enter Number of Looping (N) = 50
```

```
CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 83391512
Second Capture Execution time in millisecond: 83395420
Different Execution time in millisecond: 3908
Value of Sum from the Stress Test (polynomial) = 11783525
Press 'y' to continue or 'n' to exit the benchmark:
Enter Number of Looping (N) = 100
```

```
CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 84766849
Second Capture Execution time in millisecond: 84774683
Different Execution time in millisecond: 7834
Value of Sum from the Stress Test (polynomial) = 181630800
Press 'y' to continue or 'n' to exit the benchmark:
Enter Number of Looping (N) = 200
```

```
CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 84798944
Second Capture Execution time in millisecond: 84814606
Different Execution time in millisecond: 15662
Value of Sum from the Stress Test (polynomial) = 2852516600
Press 'y' to continue or 'n' to exit the benchmark:
Enter Number of Looping (N) = 500
```

```
CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 84841184
```

```
CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 84841184
Second Capture Execution time in millisecond: 84880270
Different Execution time in millisecond: 39086
Value of Sum from the Stress Test (polynomial) = 2816521600
Press 'y' to continue or 'n' to exit the benchmark:
Enter Number of Looping (N) = 700
```

```
CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 85602940
Second Capture Execution time in millisecond: 85657673
Different Execution time in millisecond: 54733
Value of Sum from the Stress Test (polynomial) = 1503975592
Press 'y' to continue or 'n' to exit the benchmark:
Enter Number of Looping (N) = 1000
```

```
CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 85667403
Second Capture Execution time in millisecond: 85745598
Different Execution time in millisecond: 78195
Value of Sum from the Stress Test (polynomial) = 4166126232
Press 'y' to continue or 'n' to exit the benchmark: |
```

Acer Nitro 5 Intel

```
Welcome to CPU Benchmark Program
Benchmark CPU time Using Equation  $y = 7 \times x^3 + 9 \times x^2 + 13 \times x + 25$ 
(with delay coef1, coef2, coef3, coef4 = 7, 9, 13, 25 msec)
Enter Number of Looping (N) = 50
```

```
CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 66080597
Second Capture Execution time in millisecond: 66084508
Different Execution time in millisecond: 3911
Value of Sum from the Stress Test (polynomial) = 11783525
Press 'y' to continue or 'n' to exit the benchmark:
Enter Number of Looping (N) = 100
```

```
CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 66089916
Second Capture Execution time in millisecond: 66097735
Different Execution time in millisecond: 7819
Value of Sum from the Stress Test (polynomial) = 181630800
Press 'y' to continue or 'n' to exit the benchmark:
Enter Number of Looping (N) = 200
```

```
CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 66104336
Second Capture Execution time in millisecond: 66119942
Different Execution time in millisecond: 15606
Value of Sum from the Stress Test (polynomial) = 2852516600
Press 'y' to continue or 'n' to exit the benchmark:
Enter Number of Looping (N) = 500
```

```
CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 66132494
Second Capture Execution time in millisecond: 66171526
Different Execution time in millisecond: 39032
Value of Sum from the Stress Test (polynomial) = 2816521600
Press 'y' to continue or 'n' to exit the benchmark:
Enter Number of Looping (N) = 700
```

```
CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 66179382
Second Capture Execution time in millisecond: 66234070
Different Execution time in millisecond: 54688
Value of Sum from the Stress Test (polynomial) = 1503975592
```

```
Enter Number of Looping (N) = 1000
```

```
CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 66278074
Second Capture Execution time in millisecond: 66356092
Different Execution time in millisecond: 78018
Value of Sum from the Stress Test (polynomial) = 4166126232
Press 'y' to continue or 'n' to exit the benchmark: |
```

Acer Nitro AN515-55

```
Welcome to CPU Benchmark Program
Benchmark CPU time Using Equation  $y = 7 \times x^3 + 9 \times x^2 + 13 \times x + 25$ 
(with delay coef1, coef2, coef3, coef4 = 7, 9, 13, 25 msec)
Enter Number of Looping (N) = 50
```

```
CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 71767460
Second Capture Execution time in millisecond: 71771342
Different Execution time in millisecond: 3882
Value of Sum from the Stress Test (polynomial) = 11783525
Press 'y' to continue or 'n' to exit the benchmark:
Enter Number of Looping (N) = 100
```

```
CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 71773303
Second Capture Execution time in millisecond: 71781081
Different Execution time in millisecond: 7778
Value of Sum from the Stress Test (polynomial) = 181630800
Press 'y' to continue or 'n' to exit the benchmark:
```

```
Enter Number of Looping (N) = 200
```

```
CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 71812816
Second Capture Execution time in millisecond: 71828358
Different Execution time in millisecond: 15542
Value of Sum from the Stress Test (polynomial) = 2852516600
Press 'y' to continue or 'n' to exit the benchmark:
Enter Number of Looping (N) = 500
```

```
CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 71835721
Second Capture Execution time in millisecond: 71874592
Different Execution time in millisecond: 38871
Value of Sum from the Stress Test (polynomial) = 2816521600
Press 'y' to continue or 'n' to exit the benchmark:
```

```
Enter Number of Looping (N) = 700
```

```
CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 71898915
Second Capture Execution time in millisecond: 71953252
Different Execution time in millisecond: 54337
Value of Sum from the Stress Test (polynomial) = 1503975592
Press 'y' to continue or 'n' to exit the benchmark:
Enter Number of Looping (N) = 1000
```

```
CPU time Stress Test in progress...
Result:
First Capture Execution time in millisecond: 71956263
Second Capture Execution time in millisecond: 72033966
Different Execution time in millisecond: 77703
Value of Sum from the Stress Test (polynomial) = 4166126232
Press 'y' to continue or 'n' to exit the benchmark:
```

4.0 Analysis, Comparison and Discussion

4.1 CPU-Z

Computer	Lenovo IdeaPad Flex 5	ASUS VivoBook S14	Acer Nitro 5 Intel	Acer Nitro AN515-55
CPU				
Processor	Intel Core i5 1135G7	Intel Core i5 8250U	Intel Core i7 11800H	Intel Core i7 10750H
Code Name	Tiger Lake-U	Kaby Lake-R	Tiger Lake	Comet Lake
Max TDP	28.0 W	15.0 W	45.0 W	45.0W
CPU Technology	10 nm	14 nm	10 nm	14 nm
Base Frequency	2.40 GHz	1.60 GHz	2.30 GHz	2.60 GHz
Multiplier	4.0 - 42.0	4.0 - 34.0	8.0 - 46.0	8.0 - 48.0
Number of cores	4	4	8	6
Number of Threads	8	8	16	12
L1 Data Cache	4 x 48KB (12-way)	4 x 32 KB (8-way)	8 x 48 KB (12-way)	6 x 32 KB (8-way)
L1 Instruction Cache	4 x 32 KB (8-way)	4 x 32 KB (8-way)	8 x 32 KB (8-way)	6 x 32 KB (8-way)
L2 Cache	4 x 1.25 MB (20-way)	4 x 256 KB (4-way)	8 x 1.25 MB (20-way)	4 x 256 KB (8-way)
L3 Cache	8 MB (8-way)	6 MB (12-way)	24 MB (12-way)	12 MB (16-way)
Motherboard				
Bus Specs	PCI-Express 4.0 (16.0 GT/s)	PCI-Express 3.0 (8.0GT/s)	PCI-Express 4.0 (16.0 GT/s)	PCI-Express 3.0 (8.0GT/s)
Memory				
Type	DDR4	DDR4	DDR4	DDR4
Size	8 GB	12 GB	16 GB	16 GB

Channel Number	Dual	Dual	Single	Dual
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Lenovo IdeaPad Flex 5 and Acer Nitro 5 Intel uses processors from the **Tiger Lake-U family and Tiger Lake family** and **operates on 10 nm technology**. Which are newer compared to the processor of ASUS VivoBook S14 (Kaby Lake-R) and Acer Nitro AN515-55 (Comet Lake) which operates on 14 nm technology. This may result in a **better efficiency and performance in Lenovo IdeaPad Flex 5 and Acer Nitro 5 Intel**.

On top of that, the Core i7 processors in the Nitro models have more cores and threads compared to the Core i5 processors. This allows them to **perform better for multi-threaded applications**.

Intel Core i7 10750H has a greatest base frequency of 2.60GHz up to 4.80 GHz with Turbo Boost. The **high base frequency** in Intel Core i7 10750H allow it to **standout in demanding computational tasks and intensive workloads**. On the other hand, Intel Core i5 8250U with base frequency of 1.60 GHz and up to 3.40 GHz with Turbo Boost would be **adequate to perform for basic multitasking but may struggle in heavily threaded applications**. These are reflected in the stress test using **Geekbench**.

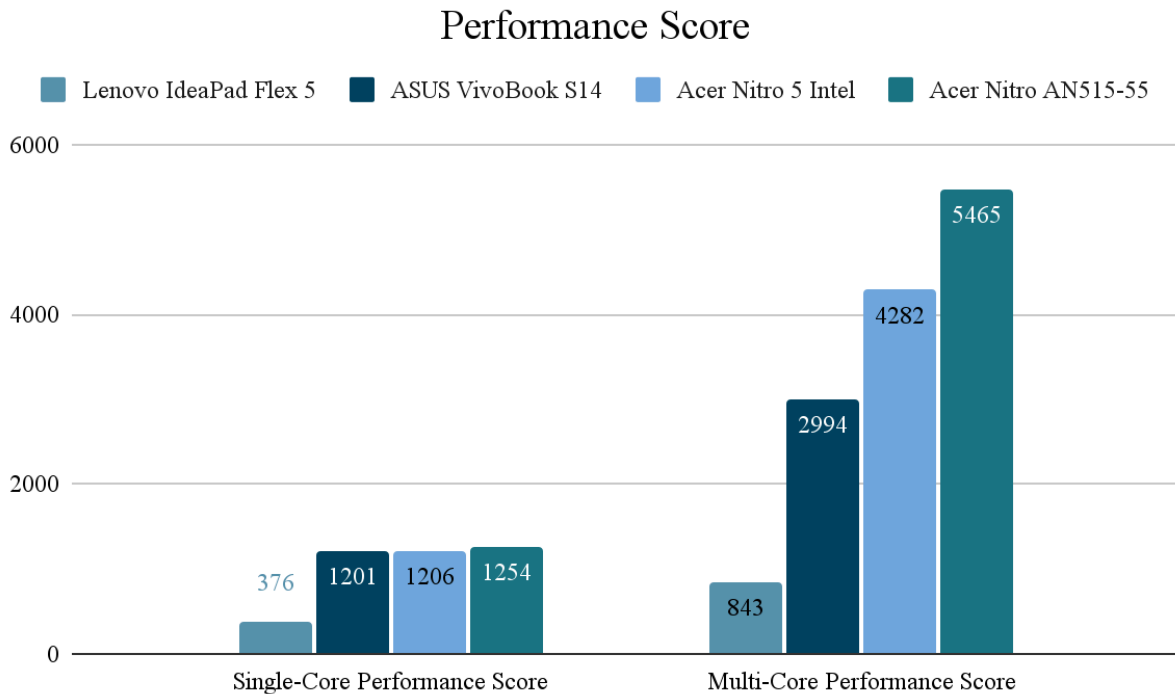
In terms of cache sizes, Intel Core i7 11800H has the largest cache. This would offer better performance in tasks that benefit from large caches. Besides, the larger cache can increase hit rate and thus increases speed.

All computers are using the PCI-Express (Peripheral Component Interconnect Express) that connects components such as graphics cards, SSDs, and other peripherals to the motherboard. Lenovo IdeaPad Flex 5 and Acer Nitro 5 Intel are equipped with PCIe 4.0 which offers double the bandwidth of PCIe 3.0. This can **enhance performance for graphics-intensive applications and high-speed storage devices**. Although PCIe 3.0 is older compared to PCIe 4.0, it is still substantial for most applications and peripherals currently available. This would be reflected in the stress test using **MiniTools**.

In terms of memory, all the four computers are equipped with DDR4 memory type which is the latest generation of Double Data Rate (DDR) Random Access Memory (RAM) that ensures the compatibility with modern applications and systems. In terms of memory capacity, Acer Nitro 5 Intel and Acer Nitro AN515-55 has the **highest memory capacity**. This would be **beneficial for running memory-intensive applications and multitasking**. Despite smaller memory capacity, Lenovo IdeaPad Flex 5 and ASUS VivoBook S14 has **dual-channel configuration** that enables the CPU to access two memory modules simultaneously, **enhancing memory bandwidth and system performance**. Conversely, the **single-channel configuration** in Acer Nitro 5 Intel may **limit peak memory bandwidth** compared to dual-channel configuration but it is still substantial

as it has 16 GB RAM. These are reflected in the memory consumption as shown in the **Task Manager**.

4.2 Geekbench



Based on the performance score from **Geekbench**, we can see that Acer Nitro AN515-55 has a slightly higher single-core performance score, suggesting a better performance in single-threaded tasks. This may be due to its **higher base frequency**.

Acer Nitro AN515-55 has the highest multi-core performance score, followed by Acer Nitro 5 Intel. This may be due to the higher-end processors with more cores. However, Acer Nitro AN515-55 outperforms Acer Nitro 5 Intel, even though it has only fewer cores and threads. This may also be due to its higher base clock speeds or other architectural advantages.

Although Lenovo IdeaPad Flex 5 and ASUS VivoBook S14 has **equal number of cores**, ASUS VivoBook S14 outperforms the Lenovo IdeaPad Flex 5.

Overall, we can conclude that Acer Nitro AN515-55 has the better and holistic performance in both single-core and multi-core tasks.

4.3 Task Manager

Computer	Lenovo IdeaPad Flex 5	ASUS VivoBook S14	Acer Nitro 5 Intel	Acer Nitro AN515-55
Low Load (30% - 50%)	Microsoft Word Task Manager	Task Manager Google Chrome	Task Manager File Explorer	Task Manager Microsoft Word Snipping Tool Brave Browser
	167.3 MB	530 MB	339.1 MB	1752.3 MB
	60%	47%	58%	48%
Middle Load (50% - 60%)	Microsoft Word Task Manager Visual Studio	Task Manager Google Chrome WhatsApp	Task Manager Opera GX	Task Manager Words Snipping Tool Brave Browser Telegram Burp Suite
	287.5 MB	1822.1 MB	1374.9 MB	1778.5 MB
	67%	59%	65%	47%
High Load (> 70%)	Task Manager File Explorer Microsoft Edge	Task Manager Google Chrome Adobe Acrobat Microsoft Excel Word WhatsApp File Explorer Filmora	Task Manager Opera GX File Explorer	Task Manager Microsoft Words Snipping Tool Brave Browser Telegram Burp Suite Firefox Google Chrome Microsoft Edge Netflix VS Code VM Player WhatsApp File Explorer
	2977.2 MB	2406.9 MB	2537 MB	4510.3 MB
	87%	72%	73%	75%

According to the information, it is considered low load if the memory consumption is less than 50%. Even with two apps running, Lenovo IdeaPad Flex 5 and Acer Nitro 5 Intel can hardly maintain a memory consumption less than 50%. However, we acknowledge that there may be some background processes or window processes that may be taking up memory.

The Lenovo IdeaPad Flex 5, with 8 GB of RAM in a dual-channel configuration, when utilising applications such as web browsing, the **system approaches its memory limits** (87%). This would **cause slowdowns** due to swapping data between RAM and storage of page files.

The ASUS VivoBook S14, with 12 GB RAM in dual-channel configuration shows better multitasking capability compared to Lenovo IdeaPad Flex 5. Even under heavy loads, including multiple browser tabs, productivity software and video editing software, the computer **can maintain its performance** with 72% memory usage.

When compared to Acer Nitro 5 Intel, despite its single-channel configuration, the system can **handle tasks effectively without hitting critical memory limits** (78%) with intensive usage including multiple tabs in the web browser.

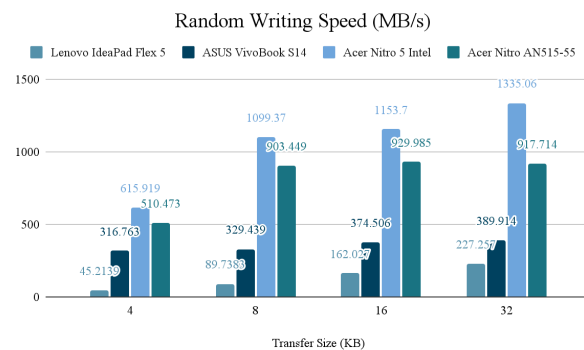
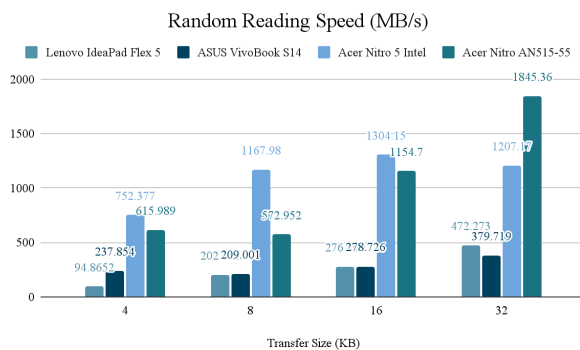
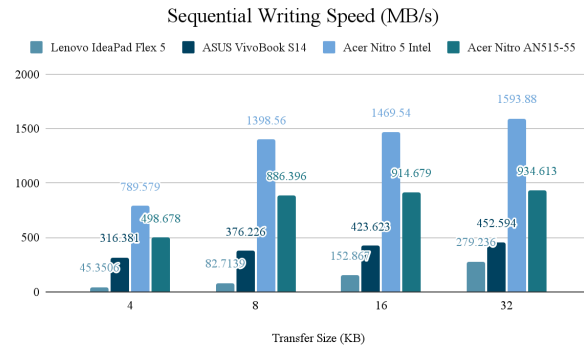
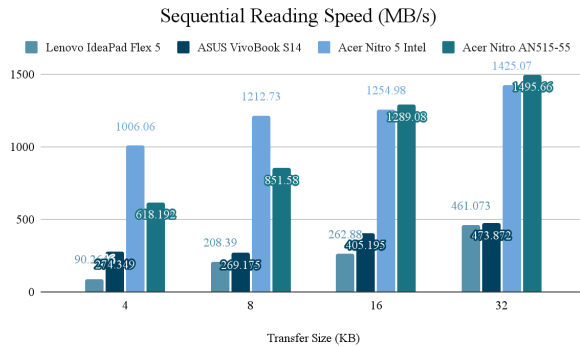
Acer Nitro AN515-55 seems to be performing the best among the four computers. It manages to maintain its performance with 75% memory usage even though various applications are running with a 4510.3MB memory consumption aside from the other background processes.

Generally, **dual-channel configurations offer better memory bandwidth and multitasking performance** compared to single-channel configuration. This can be evident by comparing the two Acer computers that have the same amount of RAM. Besides, the **larger RAM sizes allow the computer to handle more applications and browser tabs simultaneously** without running out of memory. This can be evidenced by comparing the other computer with Lenovo IdeaPad Flex 5.

4.4 MiniTools

All Test are done on respective C drive with with the following settings:

- Transfer size: 4KB to 32KB
- Queue number: 8
- Thread number: 4
- Cool down time: 5 seconds
- Total length: 200 MB



Acer Nitro 5 Intel consistently shows higher sequential and random read and write speed compared to other computers. This may be due to its newer version of PCI Express, leading to faster transfer rate. The performance is followed by Acer Nitro AN515-55, which outperformed Acer Nitro 5 Intel in reading large transfer sizes.

Lenovo IdeaPad Flex 5 and ASUS VivoBook S14 may not be as competitive as the other two computers when it comes to storage speed. This may impact tasks involving large data transfers or intensive disk operations.

4.5 ASM Code Program

	CPU_msec (different)			
Computer	Lenovo IdeaPad Flex 5	ASUS VivoBook S14	Acer Nitro 5 Intel	Acer Nitro AN515-55
50	5711	3908	3911	3882
100	7905	7834	7819	7778
200	15494	15662	15606	15542
500	38837	39086	39032	38871
700	54428	54733	54688	54337
1000	77634	78195	78018	77703

In general, as the number of loops increases, the execution time increases. We can observe that the execution times are relatively consistent with slight variations between computers.

Based on the time taken to execute the loop in the program, we can see that Acer Nitro AN515-55 uses the least time overall. This may be due to its higher base frequency.

However, as the loop number increases to 200, 500 and more, Lenovo IdeaPad Flex 5 seems to outperform other computers with the least time taken to execute the loop.

The difference in execution times between computers can be due to variations in CPU architecture, clock speeds or efficiency in handling the assembly code.

5.0 Conclusion

In conclusion, our project successfully showcased the differences in performance among various CPUs using easy-to-access benchmarking tools. By using CPU-Z, Geekbench, Windows Task Manager, and MiniTool Partition Wizard, we were able to see how each CPU handles different tasks and how well they perform. We noticed that CPUs with more cores and threads, like the Intel Core i7-10750H and Intel Core i7-11800H, performed better, especially with multi-threaded applications. This project also highlighted how important factors like cache size and PCI-Express versions are for overall performance.

Additionally, we developed a simple ASM code program to measure execution time, which added a practical element to our study. This hands-on approach helped us understand the real-world applications of benchmarking tools and techniques. Overall, we achieved our goals by learning more about CPU benchmarking, comparing different CPUs, and seeing how they handle various workloads. This experience has provided us with valuable insights that we can use in future projects and research in computer performance evaluation.

6.0 References

- Allan, P. (2014, November 25). *The differences between synthetic, real world, and hybrid benchmarks*. Lifehacker.
<https://lifehacker.com/the-differences-between-synthetic-real-world-and-hybr-1663435313>
- Different types of benchmarks*. (n.d.).
<https://www.cs.umd.edu/~meesh/cmsc411/website/projects/morebenchmarks/types.html>
- Ionescu, S. (2022, August 29). *MiniTool Partition Wizard review*. TechRadar.
<https://www.techradar.com/reviews/minitool-partition-wizard>

7.0 Appendices

7.1 Tasks Distribution

Tasks	Person In Charge
Benchmarking - CPU Z - GeekBench - Task Manager - Minitools - ASM Code	All
Assembly code	Omar Knan
Report (1.0 Introduction)	Tan
Report (2.0 Flowchart)	Knan
Report (3.0 Benchmarking result)	All
Report (4.0 Analysis comparison discussion)	Lam
Report (5.0 Conclusion)	Omar
Report (Appendix and Reference)	All
Benchmarking Video	Lam
Presentation & Reflection Video	All
Slides	All
Video Editing	Tan

7.2 Group Pictures

